AI DRIVE-IN
URBAN DIGITAL TOWNS
FOR
TRAFFIC AND EMERGENCY MANAGEMENT SYSTEM

SMARTER ROADS • SAFER CITIES • BETTER TOMORROW

AI & PREDICTION
Predicts congestion and optimizes traffic flow

DIGITAL TWIN
Real-time virtual model of the city for smart decisions

REAL-TIME MONITORING
Live data from cameras, sensors and IoT devices

SMART SIGNAL CONTROL
AI optimizes signal timing dynamically

EMERGENCY MANAGEMENT
Fastest route and priority for emergencies

VIOLATION DETECTION
Automated detection and smart enforcement

REPORTS & ANALYTICS
Data-driven insights and performance reports

NOTIFICATIONS
Real-time alerts and notifications to all stakeholders















KEY COMPONENTS OF THE SYSTEM

AI ENGINE · DIGITAL TWIN MODEL · ROAD DEVICES · TRAFFIC SIGNALS · ROUTE OPTIMIZATION · EMERGENCY SERVICES · USERS & AUTHORITIES

SAFE ROADS
Reduce accidents and save lives

FAST RESPONSE
Quick emergency response time

SUSTAINABLE CITY
Less congestion, less pollution

BETTER FUTURE
Smarter solutions for better communities

AI-Driven Urban Digital Twin for Traffic and Emergency Management System

Team members :

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Mohamed Bahaa El-Hadary

Riyad Ayman Shoura

A project related to the Software Engineering course

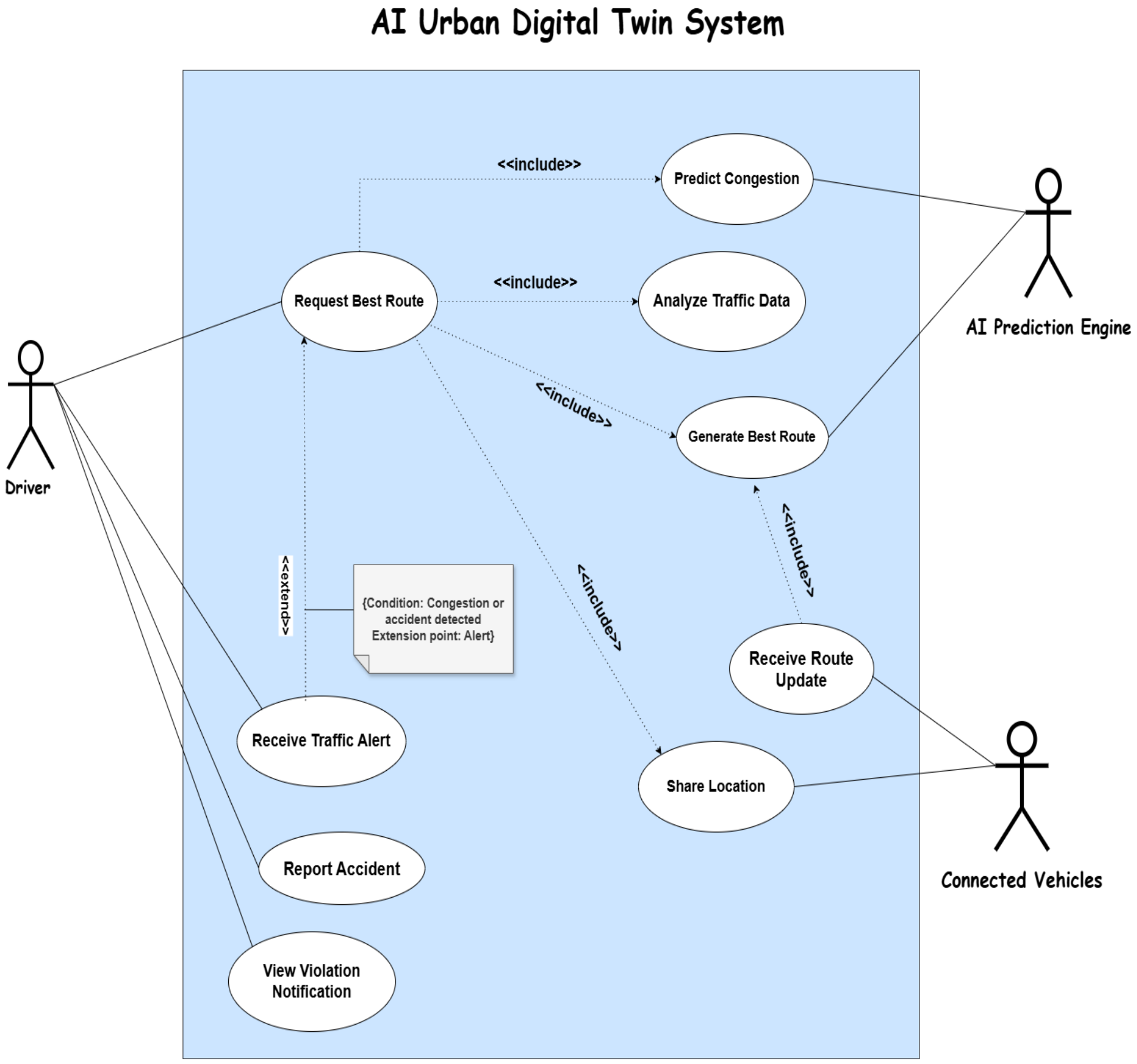
Second semester 2026

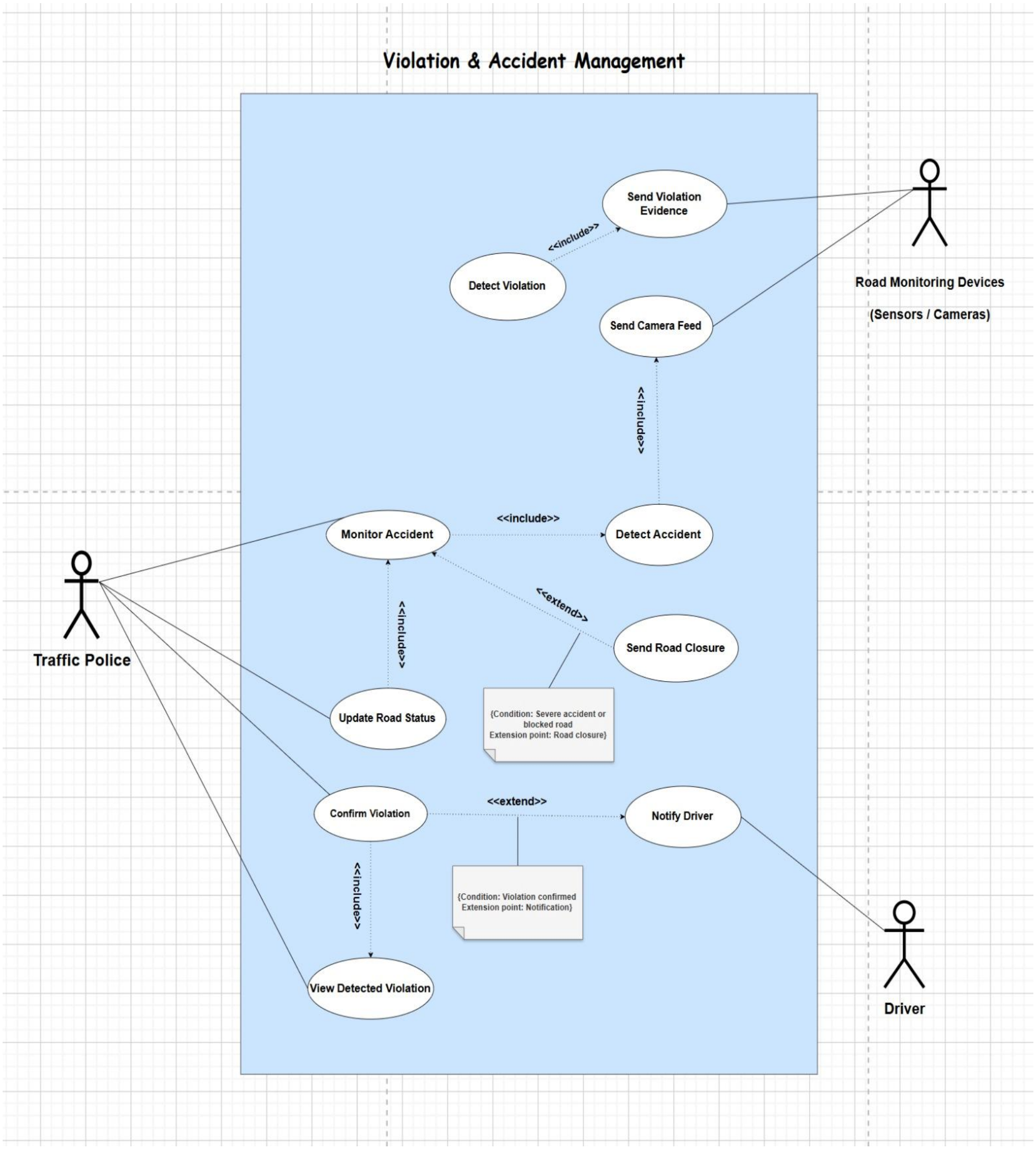
Under supervision :

Dr. Mariana Barsoum

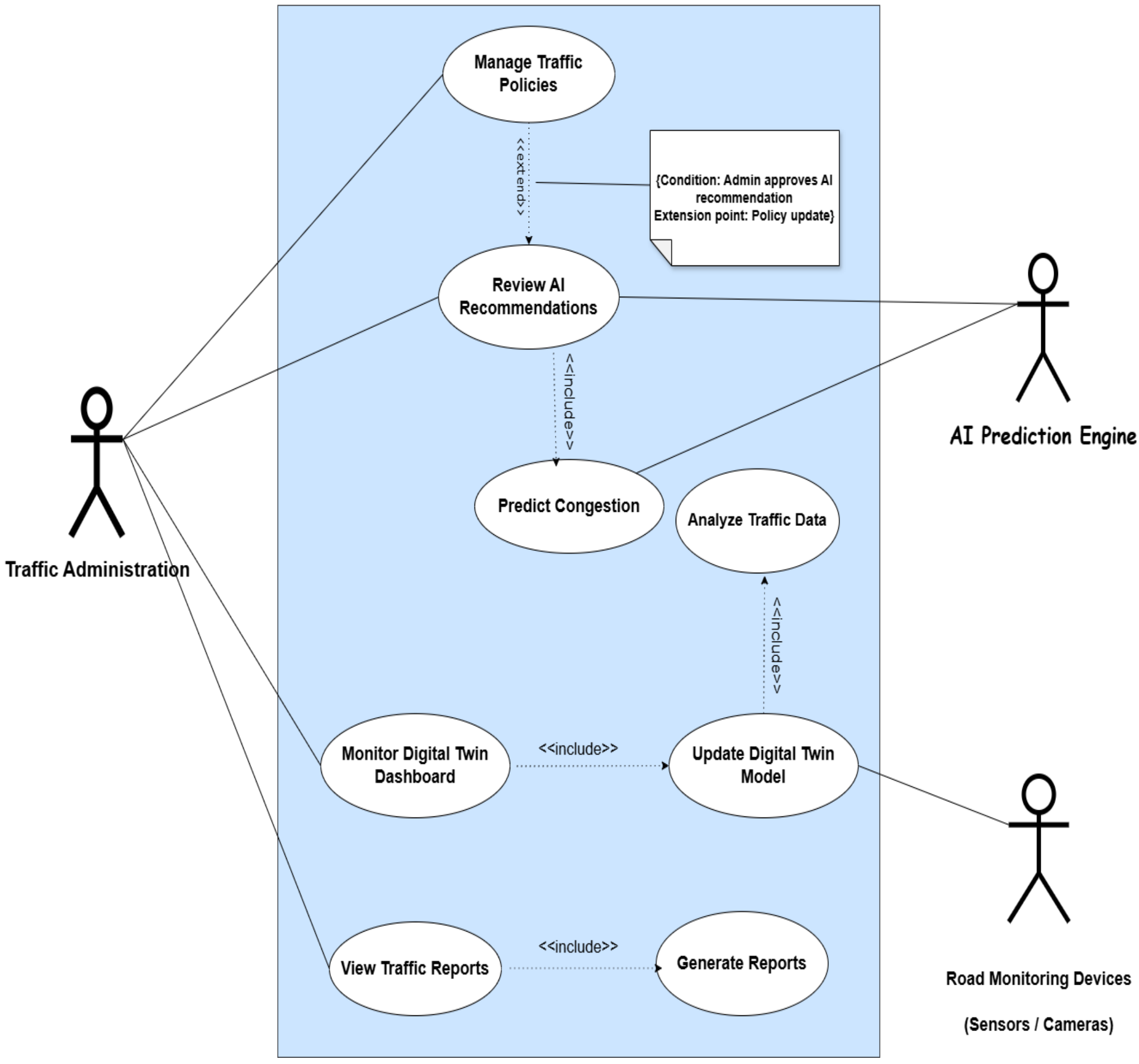
Dr. Ola Mohamed

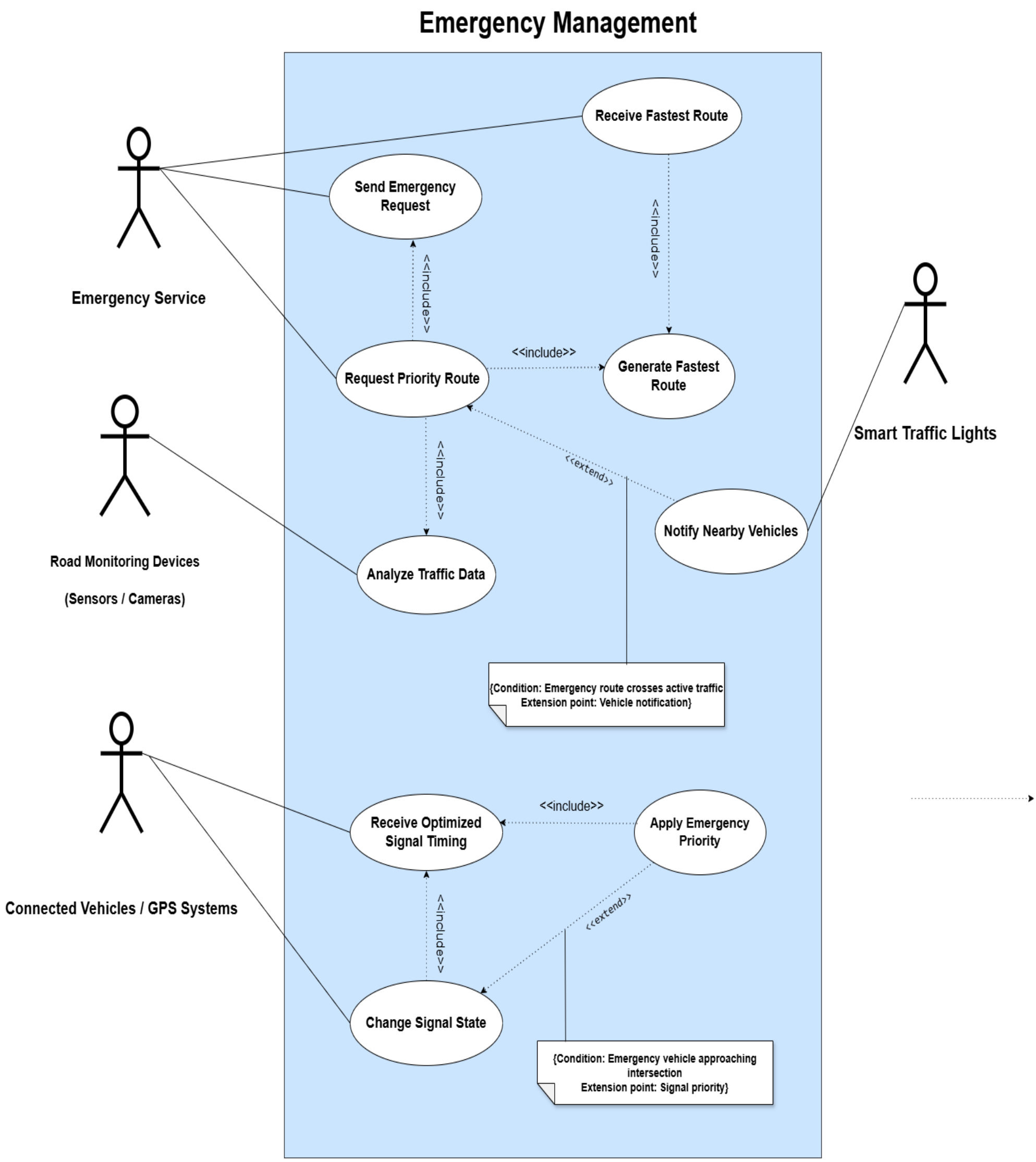
UseCase diagram



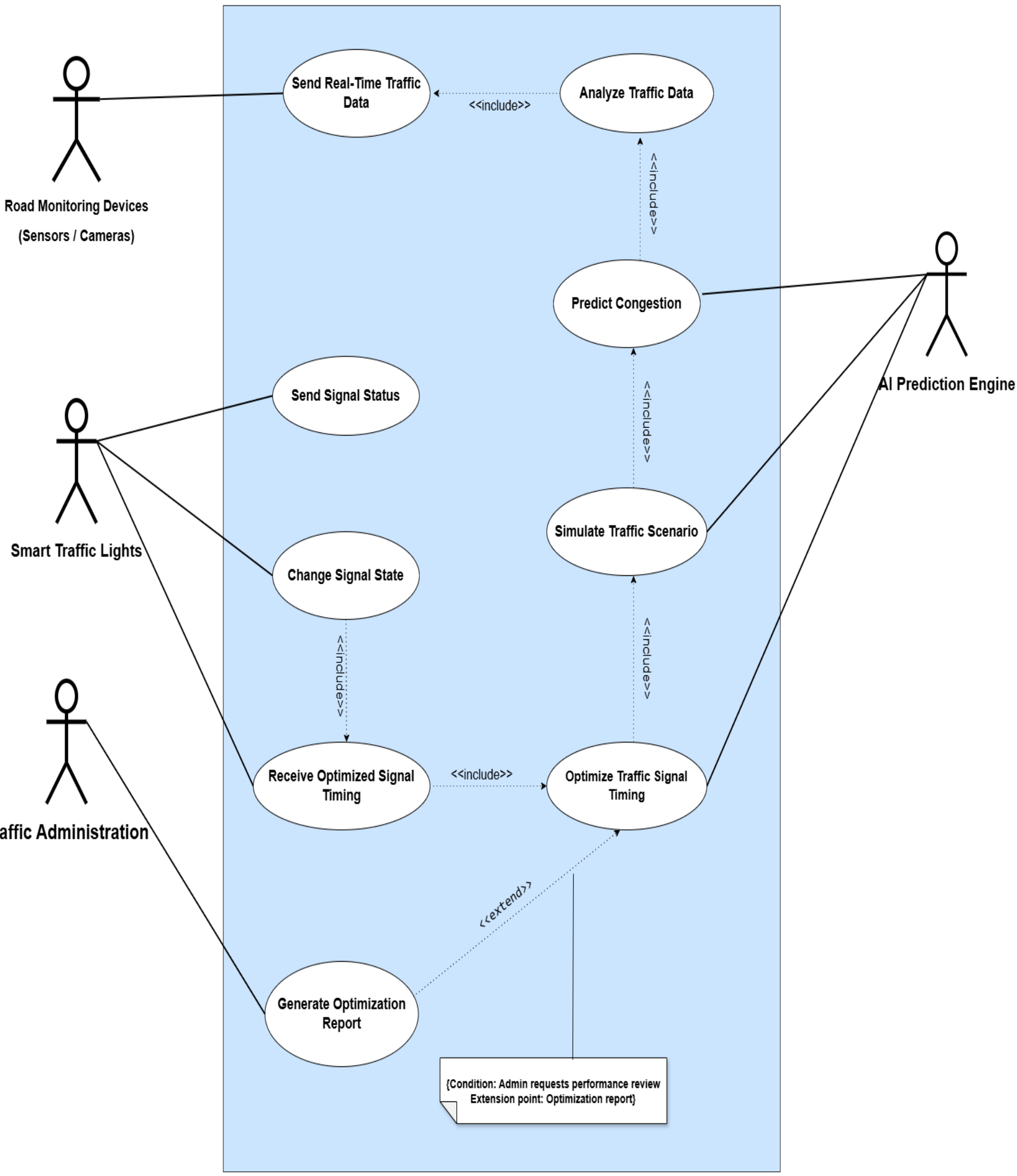

Violation & Accident Management
Send Violation Evidence
<<include>>
Detect Violation
Send Camera Feed
<<include>>
Road Monitoring Devices
(Sensors / Cameras)
Monitor Accident
<<include>>
Detect Accident
Traffic Police
<<extend>>
<<include>>
Send Road Closure
Update Road Status
{Condition: Severe accident or blocked road
Extension point: Road closure}
Confirm Violation
<<extend>>
Notify Driver
<<include>>
{Condition: Violation confirmed
Extension point: Notification}
View Detected Violation
Driver

Traffic Administration & Digital Twin Dashboard

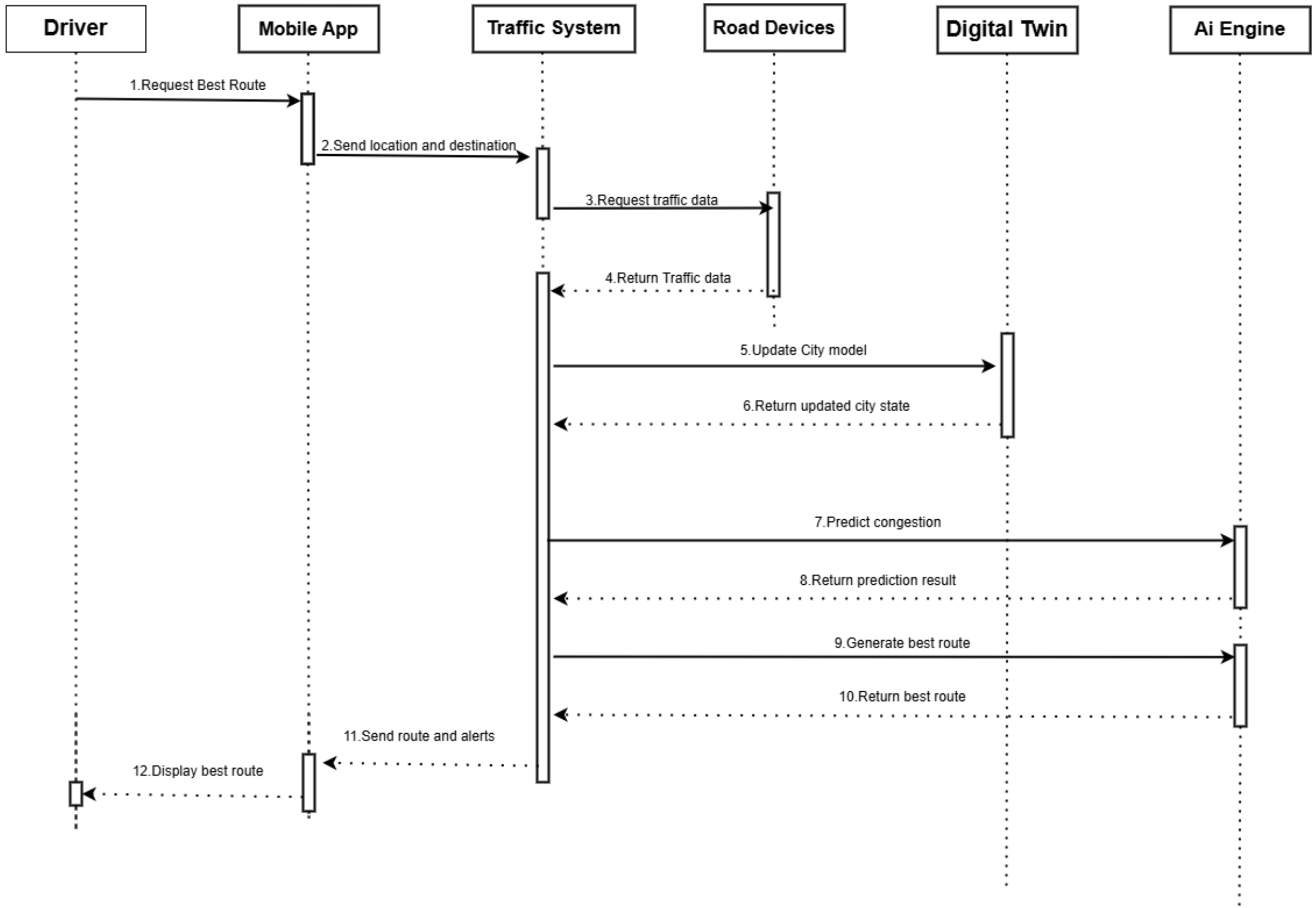



Emergency Management
Receive Fastest Route
Send Emergency Request
<<include>>
<<include>>
Emergency Service
Request Priority Route
<<include>>
Generate Fastest Route
Smart Traffic Lights
<<include>>
<<extend>>
Road Monitoring Devices
(Sensors / Cameras)
Notify Nearby Vehicles
Analyze Traffic Data
{Condition: Emergency route crosses active traffic
Extension point: Vehicle notification}
Receive Optimized Signal Timing
<<include>>
Apply Emergency Priority
Connected Vehicles / GPS Systems
<<include>>
<<extend>>
Change Signal State
{Condition: Emergency vehicle approaching intersection
Extension point: Signal priority}

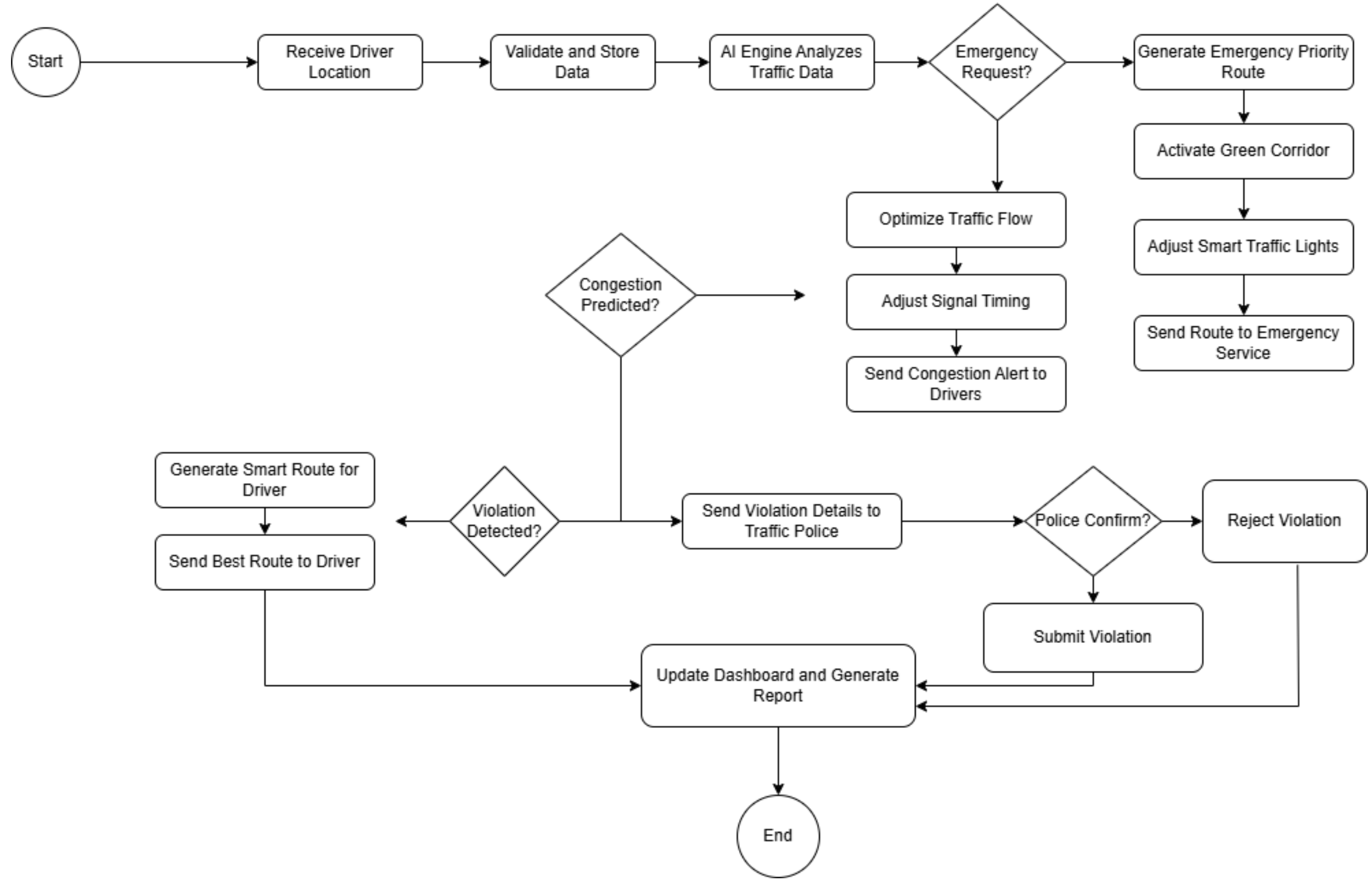
Smart Signal & AI Optimization Use Case Diagram



Sequence Diagram



Activity diagram



Class Diagram – AI-Driven Urban Digital Twin for Traffic and Emergency Management System

